

Shantanu Joshi

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EDUCATION

M.S. Software Engineering, San Jose State University, CA, USA

May 2026

Coursework: Enterprise Software Platforms, System Software Engineering, Deep Learning, Data Mining, Software Security

B.E. Computer Engineering, St. Vincent Pallotti College of Engineering, India

May 2020

Coursework: Data Structures and Algorithms, Database Management Systems, Computer Networks, Operating Systems, ML, OOP

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL

AI/ML: LangChain, Multi-Agent Orchestration, RAG Architecture, Vector Databases, LLM Integration, TensorFlow, Scikit-learn, Pandas, NumPy, NLP, PyTorch, Computer Vision, YOLOv11

Infrastructure: AWS (EC2, S3, Lambda, RDS, CloudWatch), Docker, Kubernetes, Jenkins, Kafka, Prometheus, OAuth 2.0, Microservices, CI/CD

Frameworks/Databases: React, NextJS, Spring Boot, Flask, Django, Express.js, GraphQL, REST APIs, PostgreSQL, MongoDB, Oracle, Redis, Git

RELEVANT EXPERIENCE

Full Stack Developer Intern, Corpay, USA

June 2025 - August 2025

- Rewrote 3 Express.js services across several interdependent payment flows to integrate newly deployed internal APIs while preserving backward compatibility and redesigned the React components consuming them, shipping the migration without disrupting workflows for 100K+ fleet users
- Added input validation, structured error handling, and integration tests across the rewritten endpoints, catching 2 data-contract mismatches before production, while documenting the cross-service dependencies and data contracts

Software Developer, Daxaplus Limited, Remote

June 2023 - May 2024

- Built an AI-powered grading platform with Python, Flask, and React using **GPT-4** for creative writing assessment and structured feedback generation, delivering **25% productivity improvement** across **2,000+ monthly evaluations**
- Engineered a **RAG pipeline** with **LangChain**, **Qdrant** vector storage, and **OpenAI embeddings** for semantic retrieval of grading rubrics and exemplar responses, applying prompt engineering and context window optimization to improve evaluation consistency
- Designed **PostgreSQL schemas** for submissions and grading history with JWT-based authentication, and configured **Azure DevOps CI/CD** pipelines with YAML, automated test suites, and **Docker** builds for reliable production deployment

Systems Engineer, Tata Consultancy Services, India

November 2020 - May 2023

Client: Royal Bank of Canada (RBC)

- Led migration of legacy Struts-based cheque processing to **12 microservices** using **Domain-Driven Design** for bounded context decomposition, authored OpenAPI specs, mentored 4 engineers, and cut processing time by **40%** with zero downtime for **200K+ customers**
- Engineered cross-platform **MFA APIs** with **Spring Boot** and Apigee using **OAuth 2.0** for SMS OTP, grid password, and token-based login flows, with Redis-backed session management and **JUnit** test coverage, achieving 35% faster data retrieval across mobile and web clients
- Owned the Email Automation platform built on Spring Boot **REST** and **SOAP** services, integrating **Salesforce Marketing Cloud APIs** for dynamic policy template selection and regional terms, reducing manual processing time by **95% with sub-0.5% error rate**
- Built **tokenization** services on **J2EE** and **IBM WebSphere** using JSPs, **servlets**, and SQLJ queries to secure sensitive customer data across banking workflows, implementing custom API layers aligned with **OWASP standards** across legacy mainframe systems
- Refactored **Oracle DB schemas** from inefficient join tables into master-detail relationships with three-tier operational modes, redesigned the **Angular Advisor-Portal** with state-management libraries, and streamlined frontend-middleware data flow

ACADEMIC PROJECTS

RecruitTech - AI-Powered Technical Recruitment Platform

Tech: Python, OpenAI (GPT-4o, Whisper), CrewAI, Apache Airflow, Apollo GraphQL, React, Node.js, MongoDB, Kafka, AWS S3, WebRTC, gRPC

End-to-end AI recruitment platform with separate candidate and recruiter portals: candidates apply and take live AI video interviews, recruiters get a ranked, scored shortlist. Built as **8 containerized microservices** coordinated through **Kafka** events and **gRPC**.

- Built a multi-agent screening pipeline on **Airflow** and **Kafka**, running parallel agents that score a candidate's resume, GitHub, and LeetCode profile, then a **CrewAI** agent that consolidates them into one report scored across 8 dimensions with strength and concern tags
- Built the live interview as a separate **WebRTC** microservice reached over **gRPC**, with **Whisper** for real-time transcription and GPT-4o generating and scoring up to 10 tailored questions per candidate

Thermal and RGB Fusion for Object Detection

Object detection that fuses regular and thermal camera input to stay reliable in low-light and poor-visibility conditions where RGB alone fails.

- Built and compared **three RGB-thermal fusion** strategies (early, GAN-based, and lighting-adaptive late fusion) for a **YOLOv11** detector, where the **GAN** approach learned optimal channel fusion and reached the best low-light detection at **0.755 mAP@50**

CERTIFICATIONS & ACHIEVEMENTS

NSD Certified in **Penetration Testing and Vulnerability Analysis** (ISAC India)

Won 1st place for **Most Innovative Use of Agents** at the Harness Engineering Hack for "Magpie", a multi-agent ambient study tool

Top 5 at Alien.org Hackathon for "Bingo with Aliens", a verified identity gaming platform with real-time multiplayer payment jackpot

Won Cybersecurity Track at San Jose State University Hackathon for a community-based mental health application